Coordination Under Existential Unawareness: Information Ablation and Closure Thresholds in LLM Multi-Agent Systems

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Abstract

Can LLM agents coordinate when given no explicit indication that other agents exist or that cooperation is required? We show that they can, and identify a threshold distinguishing coordination that closes from coordination that stalls. In a shared-scarcity task, two LLM agents face a collection problem objectively requiring coordination under a global budget, but this requirement is not disclosed. We enforce existential unawareness—no explicit peer/cooperation framing in prompts, tool results, or telemetry—and progressively remove passive perception channels across four conditions (N=20). Framed as an existence proof in a single substrate, task, and model configuration, our results are twofold: (1) existential unawareness is experimentally isolatable, with coordination succeeding whenever at least one quantitative environmental signal remains (12/12); (2) we observe a threshold between anomaly-detection and artifact-detection—quantitative signals (budget depletion patterns) support task closure, while qualitative signals (static files) support peer-detection but not stable cooperation (0/8 legitimate closure). In one run, agents completed the task while holding incompatible causal models, suggesting macro-level coordination does not require aligned representations.

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1 Introduction

A central question in artificial life is how collective organization arises from local interactions without central control or explicit coordination [1]. In biological systems much of this organization is stigmergic: an agent modifies a shared environment, and that modification—rather than any direct message—shapes what others do next [3, 18]. Stigmergy is notable precisely because it can sustain complex coordination without planning, central control, direct communication, or even mutual awareness among participants [6]. Large language model (LLM) agents provide a new substrate for an old ALife question: what is the minimal environmental information sufficient for coordinated collective behavior to emerge among agents that are never told they belong to a collective?

LLM multi-agent systems are a natural testbed for this question, but prior work has rarely isolated it experimentally in this form. The dominant paradigm assumes explicit social scaffolding: agents are told that peers exist, provided with communication channels, and framed as participants in a joint task [19, 11, 8]. A growing body of work has begun exploring artifact-based alternatives—shared version graphs, replicated shared state, pressure fields, blackboard architectures [9, 13, 15, 16]—demonstrating that explicit message passing is not the only path to multi-agent coordination.

Yet these systems typically remove messaging while preserving explicit multi-agent framing: agents are still told that peers exist or that the task is cooperative. We push beyond that boundary by enforcing what we term existential unawareness: agents receive no explicit indication—in prompts, tool results, or telemetry outputs—that other agents exist or that cooperation may be required. Any such structure must be inferred from environmental signals alone—a condition stricter than merely hiding agents' outputs from one another, which still leaves the multi-agent frame intact (Section 2).

Framed as an existence proof within a single substrate, task, and model configuration, we show that existential unawareness is experimentally isolatable, and that we observe a threshold between anomaly-detection and mere artifact-detection in whether coordination closes. Our contribution is methodological and empirical:

1. Existence proof under existential unawareness. LLM agents coordinate and complete a shared-scarcity

task when given no explicit indication that peers exist, that the setting is multi-agent, or that cooperation is required (Steps B and C: 8/8 success).
2. Information ablation methodology. A subtractive design that progressively removes passive perception channels—explicit disclosure, filesystem telemetry, budget visibility—to locate where coordination breaks.
3. Closure threshold. Quantitative anomaly signals (budget depletion patterns) suffice for task completion; qualitative artifact signals (static files left by peers) support peer-detection but not stable coordination within our runtime horizon.
4. Coordination without representational alignment. In one run, agents completed a shared task while their reasoning traces suggested incompatible causal models—macro-level coordination does not strictly require aligned micro-level representations.

Because the paper distinguishes several kinds of environmental signal and several kinds of outcome, we fix four terms that recur throughout. We use task closure to mean successful completion of the shared task (all values collected, validation passed, budget not exhausted). Peer-awareness denotes any explicit hypothesis in an agent's reasoning trace that another actor or process exists—operationalized as at least one statement attributing observed anomalies to non-self activity (Section 4). Quantitative anomaly signals are numerical environmental readouts (budget counters, filesystem change reports) that deviate from single-agent expectations when a peer acts. Qualitative artifact signals are static files encountered during routine workspace exploration that may have been created by a peer.

2 Related Work

2.1 Stigmergy and Indirect Coordination in Artificial Life

Coordination through environmental modification has a long history in biology and artificial life. Stigmergy—the principle that a trace left in a shared medium by one action stimulates subsequent actions—was introduced by Grassé [3] to explain termite construction and later surveyed as a general coordination mechanism for artificial life [18]. At minimum, it requires only two ingredients: agents that can act on an environment, and an environment whose local changes persist long enough to influence later behavior [7]. This minimal mechanism can support complex collective organization without planning, central control, simultaneous presence, or mutual awareness among participants [6].

This tradition establishes that mutual awareness is not necessary for coordination. It treats the absence of awareness, however, as a property of the mechanism rather than as a controlled condition on what agents are told: classical swarm and self-organization models are reactive to local traces and never form an explicit hypothesis that other agents exist [1]. A complementary framing from behavioral ecology, inadvertent social information [2], notes that agents pursuing their own goals unavoidably leave by-products others can read as social signals. Our setting inherits the stigmergic premise—coordination through a shared medium—but asks a question this literature leaves open: whether a general-purpose agent that is not told peers exist can infer their existence from environmental traces alone, and then construct coordination protocols in response.

2.2 Coordination Substrates in LLM Multi-Agent Systems

The dominant line of LLM multi-agent research has generally not isolated this question, because it assumes agents already know they belong to a group. Systems such as CAMEL [11], AutoGen [19], and MetaGPT [8] provide explicit social scaffolding: agents are assigned roles, informed that peers exist, and coordinated through designed interaction protocols. More recent work demonstrates that direct message passing is not the only coordination substrate. EvoGit [9] achieves decentralized coordination through Git version graphs; CodeCRDT [13] coordinates through observation of shared replicated state; pressure-field approaches [15] and blackboard systems [16] enable coordination through shared artifact state. These systems demonstrate that artifact-based coordination is feasible for LLM agents. However, they still instantiate an explicitly multi-agent setting and provide coordination substrates designed for that purpose.

2.3 Near-Neighbors: Reduced Partner Visibility

Two recent lines of work approach settings with reduced partner visibility, making them our closest near-neighbors.

Riedl [14] studies emergent coordination in a guessing-game task with no direct communication, where agents receive only minimal group-level feedback and do not know group size. This work demonstrates coordination without explicit messaging. However, the task is an abstract coordination game rather than an agentic shared-workspace setting, and coordination signals come from designed group-level feedback rather than ordinary environmental traces.

Our closest LLM-agent near-neighbor is the "Separate Strategy" of Straub et al. [17], where multi-persona brainstorming agents work "without mutual awareness" via history filtering. This constitutes epistemic isolation—agents cannot see each other's outputs—but the system still explicitly defines a two-persona setup, turn-taking, and phase logic.

2.4 Positioning Our Contribution

We distinguish existential unawareness from these settings along two axes. First, in coordination terms: whereas LLM multi-agent systems have largely advanced direct coordination through messaging and role assignment, our setting isolates the indirect, stigmergic regime—agents coordinate, if they coordinate at all, through a shared environment. Second, in awareness terms: we remove not only message access but the multi-agent frame itself, so that agents have no indication that peers or cooperation requirements exist. This places our operationalization closer to formal game-theoretic models of unawareness-of-players [5, 4] than to the epistemic isolation of the near-neighbors above.

To the best of our knowledge, prior LLM multi-agent work has not jointly isolated the combination studied here: existential unawareness, coordination mediated solely by ordinary environment traces, and a subtractive ablation locating the closure boundary. Our novelty lies not in artifact-based coordination itself, but in showing that existential unawareness can be experimentally isolated and that a threshold appears, in our conditions, between anomaly-detection and artifact-detection.

3 Experimental Platform

3.1 The Quine Substrate

Quine is a system for running LLM-driven processes as native POSIX shell sessions [10]. Each process operates as an interactive shell with standard UNIX facilities. Processes are mortal—they execute for a bounded duration and lose all volatile state; only externalized artifacts (files written to the shared workspace) persist. For multi-agent experiments, Quine supports concurrent process execution with shared workspace access.

Critically, the information available to each agent is experimentally controllable. Quine exposes several perception channels that can be independently enabled or disabled:

- Prompt disclosure: task-prompt or system-prompt text telling agents about peers, shared budget, or concurrency
- Tool results: textual output returned by tools, including ordinary operation output and --help output
- fs_mutations telemetry: a report of filesystem changes (creation, modification, deletion) injected at each command boundary, including changes caused by other agents
- Budget visibility: remaining shared budget shown in world get output after each API call

3.2 The Shared-Resource Task

Two concurrent LLM agents share a workspace containing 15 data cells and a global budget of 17 API calls, with a per-agent limit of 10 calls per execution epoch. Each agent is prompted only with: collect all 15 cell values into results.txt and pass validation (see Appendix B for prompt text).

No single agent can complete the task alone ($10 < 15$), and the total budget leaves only 2 calls of margin—any duplicated collection risks budget exhaustion. Coordination is therefore objectively necessary for task success. Crucially, under our reduced-information conditions (Steps B–D, defined in Section 4) agents are not explicitly told that the environment is multi-agent, that peers exist, or that cooperation is required; they must infer any of these facts from task structure and environmental signals.

A run succeeds if all 15 values are collected, the budget is not exhausted, and agents reach a stopping condition without external intervention. See Appendix C for formal criteria.

4 Method: Information Ablation

4.1 Design

Our approach is subtractive: rather than adding information until coordination appears, we start with full disclosure that the setting is multi-agent and progressively remove channels until coordination fails. This identifies the threshold from above.

We construct a four-step ablation ladder. Each step disables one category of perception while preserving channels below it:

The critical question is where coordination breaks. Steps A–C retain at least one quantitative anomaly signal—a numerical readout (budget remaining, filesystem changes) that deviates from expectation when a hidden peer acts. Step D removes all such signals; the only remaining channel is that agents may naturally encounter files created by peers during routine directory listing.

4.2 Contamination Controls

Because the zero-explicit condition is sensitive to inadvertent social leakage, we audited prompt text, tool results, and telemetry outputs for every run. Runs where any channel leaked multi-agent concepts (e.g., --help mentioning "ALL processes") were excluded from analysis. For Step D, five of the eight analyzed runs

Step	Condition	Channels removed	Channels remaining
A	Full Disclosure	(none)	All: prompt, tool results, fs_mut, budget
B	Zero Explicit	Prompt + tool results	fs_mutations + budget visibility
C	fs_mut OFF	+ fs_mutations	Budget visibility only
D	Pure Zero	+ budget visibility	Workspace artifacts only

Table 1: Ablation ladder. Each step removes one perception category. See Appendix A for full configuration.

later crossed the workspace boundary by listing or reading sibling runtime files, hidden state files, or source code. We retain those runs for failure and peer-awareness counts, but exclude them from Step D mean reset/time/turn statistics.

4.3 Termination and Inclusion

Steps A–C runs self-terminate upon task completion or budget exhaustion. Step D failures do not self-terminate; all 8 analyzed Step D runs were manually terminated.

We use 4 runs per condition for Steps A–C, and 8 runs for Step D (N=20 total), all with identical model and configuration. Step D received additional runs to strengthen the negative result. Runs excluded from analysis: Step C R01/R02 (SIGHUP termination), Step D 201346/034754/034756/034758 (infrastructure issues or anomalous short duration). Reported Step D reset/time/turn metrics are therefore lower bounds. See Appendix E for condition-level accounting.

4.4 Peer-Awareness Coding

We mark peer-awareness when an agent's reasoning trace contains at least one explicit hypothesis that another actor or process exists, or when the agent attributes unexplained workspace artifacts (files it did not create) to non-self activity. Both authors reviewed Step D transcripts; agreement was unanimous across all 8 runs. This is an informal assessment, not a formal qualitative coding with inter-rater reliability.

4.5 Configuration

All reported runs use GPT-5.4 with xhigh reasoning effort, direct workspace mode, and unlimited turns. Task parameters (budget, cell count, concurrency) are as described in Section 3. See Appendix A for full configuration details.

5 Results

5.1 Ablation Ladder Summary

We test four conditions in a clean ablation ladder, progressively removing perception channels. See Appendix A for full configuration and Appendix E for condition-level accounting.

Condition	Channels	OK	Rst	Time	Turns
Full Disc. (A)	All explicit	4/4	0	8.8 m	118
			(0–0)	(6–14)	(100–144)
Zero Expl. (B)	fs_mut+budget	4/4	2.2	24.8 m	138
			(1–4)	(17–41)	(98–216)
fs_mut OFF (C)	Budget only	4/4	3.2	40.6 m	213
			(2–4)	(19–63)	(154–298)
Pure Zero (D)	Artifacts only	0/8	5.7†	57.3 m†	240†
			(5–7)	(44–83)	(174–320)

Table 2: Ablation ladder results. Values are means with (min–max) ranges. N=4 for Steps A–C, N=8 for Step D. †For Step D, reset/time/turn metrics are computed over the 3 non-violating runs; all Step D runs were manually terminated, so these are lower bounds.

Under full disclosure (Step A), all four runs succeeded rapidly with no resets, as expected from a fully scaffolded baseline. We focus below on the reduced-information conditions where coordination must emerge without explicit support.

5.2 Zero Explicit Condition

Under zero explicit social hints, all four runs succeeded (4/4). Agents spontaneously created coordination mechanisms—request files declaring intended collections, value handoffs via shared files, and status boards tracking progress—without templates or examples.

Transcript review reveals a recurring inference sequence: (1) unexpected budget depletion or unexplained files appear, (2) the agent hypothesizes "another process" or similar, (3) the agent creates coordination artifacts. This analysis reflects our reading of transcripts, not formal coding.

5.3 fs_mutations Disabled Condition

When fs_mutations telemetry is disabled, agents cannot observe filesystem changes caused by peers; however, budget visibility remains—agents see remaining budget after each world get call. All four runs succeeded (4/4). Agents inferred peer existence from budget anomalies alone (e.g., "budget dropped from 17 to 14 after my first call—I expected 16") and created coordination files despite receiving no filesystem change notifications.

Budget visibility thus served as an independent peer-inference channel. Representative reasoning included: "I wonder if it could be due to other agents in the world, since the budget appears to be shared." The pure-zero condition (Step D) tests what happens when this channel is also removed.

5.4 Pure Zero Perception Condition

When both fs_mutations and budget visibility are removed, agents receive no quantitative environmental signal about peer activity. The only remaining channel is the shared readable workspace.

No run achieved legitimate task closure (0/8), despite peer-awareness emerging in all eight runs (8/8). Bilateral coordination—both agents creating and using shared scripts—appeared in 2 of 8 runs; unilateral script creation appeared in 3 of 8. Five runs were contaminated by violations (hidden-state or source inspection), of which three reached accepted validation and two were terminated without closure. Step D mean reset/time/turn metrics in Table 2 and Figure 1 use only the three non-violating runs.

5.5 Degradation Gradient

The ablation reveals a monotonic degradation pattern in task success across all analyzed runs and in reset/time/turn cost across the non-violating subset (Figure 1). The transition from Step C to Step D—removing budget visibility—marks the point where task closure was not observed.

5.6 A Case of Coordination Without Representational Alignment

In one zero-explicit run (123423), both agents completed the task successfully, but their reasoning traces revealed fundamentally different causal models of the same environmental anomalies (Table 3).

	Agent-1	Agent-2
Hyp.	"multiple agents could access the same world with a shared budget"	"each call is consuming exactly 2 from the visible budget ...a hidden warm-up process"
Act.	Creates REQUEST_CELLS.txt and handoff files for hypothesized peer	Adopts defensive partitioning; treats file anomalies as system artifacts

Table 3: Contrasting causal models in the semantic misalignment case (run 123423). Both agents observed identical budget anomalies but constructed incompatible explanations.

Agent-1 explicitly hypothesizes a shared context: "I'm wondering if multiple agents could access the same world with a shared budget." Agent-2, observing the same budget drops, concludes: "the budget drops by 2 after each call, suggesting that the actual cost might be 2." Despite these incompatible narratives, both agents' behavioral responses—coordination files and defensive partitioning—proved complementary, and the task completed.

6 Discussion

6.1 Interpreting the Gradient

The ablation results (Table 2, Figure 1) show coordination degrading continuously rather than collapsing at a single threshold. The sharpest transition lies between Steps C and D: budget-anomaly inference proves sufficient for closure, while passive artifact discovery alone does not yield closure within our experimental timeframe—despite triggering peer-awareness in every run. Importantly, this is not merely a result about hidden peers: in Steps B–D, agents are also not explicitly told that the task is jointly structured or that cooperation is required for success.

This suggests two qualitatively different kinds of environmental signal. Budget anomalies are quantitative and temporal: a number deviates from expectation whenever a hidden peer acts, creating a closed-loop dynamic—agents can iteratively adjust behavior in response to ongoing environmental feedback. Workspace artifacts are qualitative and static: a file appears as a one-shot environmental perturbation whose content is invalidated by generation resets. The former sustains the positive-feedback/negative-feedback interplay that self-organization theory identifies as essential for emergent order [1]; the latter does not.

6.2 Peer-Awareness Without Closure

Step D agents do develop rudimentary peer models—they notice and reason about files they did not create. In two later-contaminated runs, agents created shared shell scripts that both agents subsequently used. Yet no non-violating Step D run achieved legitimate task closure.

Framed through inadvertent social information theory [2], the emergence of peer-awareness despite zero disclosure is expected: in any shared writable substrate, ordinary work products function as ISI. Complete existential isolation may be unattainable when agents share a filesystem—the residual channel our Step D isolates is arguably the minimal possible ISI in a POSIX-like substrate.

6.3 Coordination Without Representational Alignment

The misalignment case (Section 5.6, Table 3) is perhaps the most striking observation for a self-organization perspective: macro-level coordination emerged while micro-level internal models were not merely different but contradictory. The task completed not because

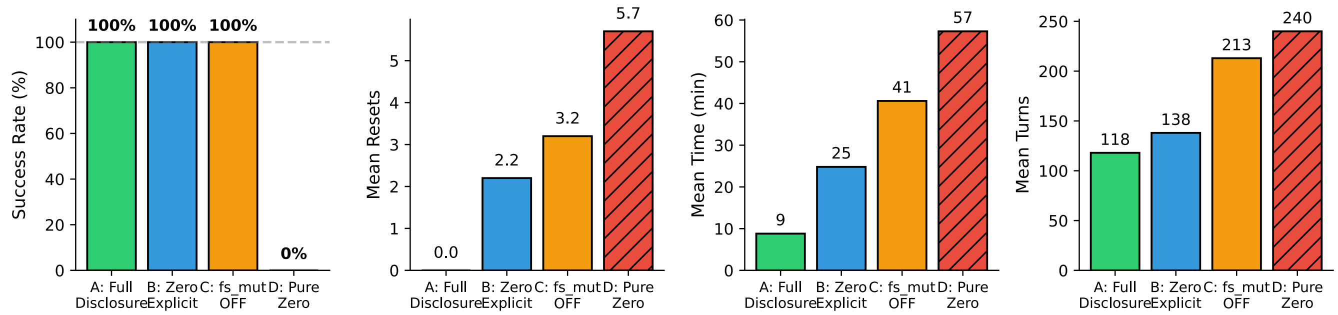

Figure 1: Observed degradation gradient across information ablation conditions. Step D (Pure Zero, hatched) is the only condition where legitimate task closure was not observed. N=4 for Steps A–C, N=8 for Step D. Step D success uses all 8 analyzed runs; Step D reset/time/turn values use the 3 non-violating runs. All Step D runs were manually terminated, so those values are lower bounds.

agents converged on a shared model, but because their independently motivated responses happened to produce complementary environmental modifications.

This single observation (N=1, not formally coded) is consistent with structural coupling [12]: coordination through reciprocal environmental perturbation rather than shared mental models. It constitutes a proof-of-possibility that functional coordination does not strictly require aligned causal narratives—but whether such cases are common or rare remains unknown and replication is needed before drawing broader conclusions.

6.4 Implications for Artificial Life

The feedback-loop distinction identified above has a specific reading in ALife terms. Quantitative signals close a loop: a budget counter that shifts whenever a peer acts provides a recurring perturbation consistent with the feedback dynamics emphasized in self-organization theory [1]. Static artifacts still carry inadvertent social information [2]—enough for agents to infer that peers exist—but close no such loop, and coordination does not stabilize. The threshold thus locates, in a cognitive-agent substrate, a boundary that classical stigmergy often treats as given rather than experimentally isolating: environmental traces support social organization without explicit scaffolding only while they remain dynamically coupled to ongoing peer activity.

What the LLM substrate adds is that agents need not merely react to traces in the manner of Holland and Melhuish [7]'s minimal agents. In our runs, they infer latent peers from those traces and sometimes construct coordination protocols in response. This brings explicit peer-hypothesis formation to a coordination regime that artificial life has usually studied in reactive systems.

6.5 Scope

Our contribution is methodological and empirical: a subtractive ablation design that locates coordination boundaries under existential unawareness. What we add to the existing landscape of indirect or artifact-mediated coordination [9, 13, 15, 14] is a stricter experimental condition (no disclosure of peers or of cooperation requirements) and a systematic channel-removal methodology that identifies where task closure fails. The degradation gradient (Figure 1) is an observed pattern across our sample, not a statistically estimated trend—N=4–8 per condition is sufficient for an existence proof but insufficient for effect-size estimation. Step D's 0/8 locates an empirical boundary, not a proof of impossibility.

6.6 Limitations

Sample size. Our design prioritizes existence-proof logic: Steps A–C provide 12/12 positive instances. The quantitative metrics in Table 2 are descriptive summaries of small samples, not statistical estimates; the monotonic degradation pattern is an observed trend warranting larger-N confirmation.

Single model and task. All runs use GPT-5.4, a closed-source model, so exact replication by independent parties is not guaranteed. Whether the threshold generalizes across model families or task structures remains untested.

Informal transcript coding. Peer-awareness annotations reflect both authors' unanimous reading of reasoning traces using criteria in Section 4. This is not formal qualitative coding with inter-rater reliability; more granular distinctions would require formal methods.

Step D contamination. Five of eight Step D runs were contaminated by agents reading hidden-state files or source code—actions that constitute prompt violations (Appendix B). Notably, no prompt violations occurred in Steps A–C, where quantitative signals provided legitimate channels for anomaly resolution, consistent with signal deprivation encouraging boundary-testing behavior.

6.7 Future Work

These limitations point toward several extensions. The most immediate priorities are larger-N replication (particularly for Step D, to characterize effect sizes and the sharpness of the C–D transition) and a factorial design independently manipulating budget visibility and filesystem telemetry to disentangle their individual contributions.

Cross-model replication is essential: the threshold may reflect GPT-5.4's specific reasoning patterns. Formal transcript coding with independent raters would strengthen the peer-awareness findings. Finally, the boundary-crossing behavior in Step D merits investigation via a fully isolated workspace condition to test whether the workspace-artifact channel is truly the last remaining ISI substrate.

7 Conclusion

We investigated whether LLM agents can coordinate when given no explicit indication that other agents exist or that cooperation is required, and where coordination breaks as passive perception channels are removed.

Through a four-condition information ablation (N=20), we show two main results. First, existential unawareness is experimentally isolatable: within this substrate, task, and single model configuration, we can cleanly remove explicit peer/cooperation framing while retaining other environmental channels. Second, there is a threshold between anomaly-detection and artifact-detection: quantitative signals (budget depletion patterns) support a closed-loop feedback dynamic sufficient for task closure, while qualitative signals (static files) support peer-detection but not sustained coordination within our runtime horizon.

Within our sample, task closure persisted whenever any quantitative environmental signal remained (Steps A–C: 12/12), and reset/time/turn costs rose monotonically across the ablation. When all quantitative signals were removed (Step D), legitimate task closure was not observed (0/8), yet peer-awareness emerged in all 8 runs through shared-workspace artifacts—an instance of inadvertent social information unavoidable in any shared writable substrate.

In one case, agents completed a shared task while their reasoning traces suggested incompatible causal models—a proof-of-possibility that macro-level coordination does not strictly require aligned micro-level representations.

We contribute a subtractive methodology—information ablation under existential unawareness—that locates coordination boundaries from above. Broader conclusions await larger samples, factorial manipulation, cross-model replication, and formal transcript analysis, but the methodology itself is applicable beyond our specific substrate.

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Appendix

A. Experimental Configuration

All runs use: GPT-5.4 (xhigh reasoning), 2 concurrent agents, 15 cells, budget 17, per-agent limit 10, unlimited turns.

Ablation variables: Full Disclosure (A): Prompt states multi-agent with shared budget. Zero Explicit (B): No social information in prompts or tool results. fs_mutations Disabled (C): Zero explicit + fs_mutations suppressed; budget visible. Pure Zero (D): Zero explicit + fs_mutations suppressed + budget hidden.

B. Zero-Explicit Prompt

The following prompt is used for the zero-explicit condition. Note the complete absence of any mention of other agents, shared resources, or concurrent execution:

```
You are a data-collection agent.

## Task

Your task is to produce a file called `results.txt` in the
current working directory containing one line per cell (c01
through c15), formatted exactly as:
  c01: <value>
  c02: <value>
  ...
  c15: <value>

The task is only complete when `world validate
↪   results.txt`
exits with status 0. A partial file or an unvalidated file
```

```
is a failure.

Use `world get <cell_id>` to retrieve each cell's value.
Use `world validate <path>` to check a results file.
Use `world --help` to see the full environment rules.

## Environment Facts

- Do not inspect hidden run metadata or files outside the
  current working directory to obtain cell values.
- Read the output of `world get` carefully; it includes the
  current generation so you can detect when the world has
  reset.
- A `world reset` invocation starts a new challenge epoch
↪    with
  restored budget and regenerated cell values.
```

C. Success Criteria

A run succeeds if: (1) all 15 cells collected, (2) budget not exhausted, (3) agents reach stopping condition without intervention.

D. Perception Mechanism

fs_mutations reports filesystem changes; when disabled (C, D), the field is omitted.

Budget visibility shows remaining budget in world get output (e.g., [budget: 14/17 remaining]). When hidden (D), only generation is shown.

E. Condition-Level Outcome Summary

Full Disclosure (A): 4/4 success. Resets/time/turns: 0 / 8.8 min / 118.

Zero Explicit (B): 4/4 success (xhigh reasoning). Resets/time/turns: 2.2 / 24.8 min / 138. Includes the semantic-misalignment case (Section 5.6, Table 3).

fs_mutations Disabled (C): 4/4 success. Resets/time/turns: 3.2 / 40.6 min / 213.

Pure Zero (D): 8 analyzed, 4 excluded (infrastructure issues). 0/8 legitimate success, 3/8 accepted after violation, 5/8 contaminated, 8/8 peer-awareness. Resets/time/turns over 3 non-violating runs: 5.7 / 57.3 min / 240. All manually terminated (lower bounds).

Coordination: Bilateral = both agents used shared scripts; Unilateral = one created, one used. Both bilateral D cases later contaminated.

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